GORAKSHA – Pashu Sanjeevini

Engineering Project Report

Web-Based Rural Animal Healthcare Platform

Cover Page

Project Title

GORAKSHA – Pashu Sanjeevini

Project Type

Web-Based Rural Animal Healthcare and Veterinary Assistance Platform

Submitted By

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Technology Stack

- HTML5
- CSS3
- JavaScript
- Claude AI API Integration
- Responsive Web Design

Academic Year

2025 – 2026

Department

Computer Science / Information Science / Artificial Intelligence & Machine Learning

Submitted To

Department of Engineering

Certificate

This is to certify that the project entitled **“GORAKSHA – Pashu Sanjeevini”** is a bonafide work carried out by the student in partial fulfillment of the requirements for the award of the engineering degree.

The project demonstrates the design and development of a rural animal healthcare platform focused on emergency veterinary assistance, AI-powered healthcare support, first aid guidance, government livestock schemes, and veterinary doctor connectivity.

The project has been completed successfully under the guidance of the project supervisor and has met the required academic standards.

Project Guide Signature: _____

Head of Department Signature: _____

Student Signature: _____

Date: _____

Acknowledgement

I express my sincere gratitude to all those who supported and guided me during the development of the project “GORAKSHA – Pashu Sanjeevini”.

I thank my project guide, faculty members, and department staff for their valuable suggestions, encouragement, and technical guidance throughout the project development process.

I also thank my family and friends for their constant support and motivation.

Special thanks to modern web technologies and AI-based development platforms that helped in building an innovative solution for rural veterinary healthcare.

This project enhanced my knowledge in frontend web development, responsive UI design, API integration, user experience design, rural problem solving, and healthcare-oriented application architecture.

Abstract

GORAKSHA – Pashu Sanjeevini is a modern web-based rural animal healthcare platform designed to assist farmers, livestock owners, and veterinary healthcare providers. The system focuses on solving major problems faced in rural veterinary healthcare such as lack of emergency support, delayed treatment, insufficient awareness about animal diseases, poor access to veterinary doctors, and lack of information regarding government livestock schemes.

The platform provides an integrated digital ecosystem where farmers can access emergency veterinary support, search nearby veterinary doctors, receive first aid guidance, interact with an AI-powered chatbot, and explore government welfare schemes related to livestock and animal healthcare.

The project is developed using HTML, CSS, and JavaScript with a mobile-responsive user interface. The system also integrates Claude AI API for intelligent chatbot assistance. The interface is designed with multilingual and farmer-friendly usability principles to support rural communities.

The platform includes multiple innovative modules such as:

- Emergency veterinary assistance system
- Veterinary doctor finder
- Offline first aid support
- AI chatbot guidance system
- Government scheme awareness portal
- User profile management
- Mobile-first responsive design
- Search and filter mechanisms
- Real-time interactive components

The application focuses heavily on accessibility, simplicity, visual communication, and emergency responsiveness.

The project demonstrates how modern web technologies and artificial intelligence can be used to improve rural livestock healthcare infrastructure.

Introduction

Livestock plays a very important role in the rural economy of India. Farmers depend on cows, buffaloes, goats, sheep, and poultry for milk production, farming activities, transportation, and income generation. Animal health directly affects agricultural productivity and rural financial stability.

However, rural veterinary healthcare systems face multiple challenges such as:

- Lack of nearby veterinary hospitals
- Delayed emergency medical support
- Poor awareness of diseases
- Lack of digital veterinary services
- Transportation difficulties
- Inadequate access to government schemes
- Limited healthcare information in local languages

In many villages, farmers lose animals due to delayed treatment and lack of first aid awareness. Many farmers also do not know how to identify symptoms during the early stages of diseases.

GORAKSHA – Pashu Sanjeevini is developed to bridge this gap between rural farmers and veterinary healthcare services.

The system acts as a digital veterinary assistant that enables farmers to:

- Find nearby veterinary doctors
- Access emergency support
- Learn first aid procedures
- Communicate with AI healthcare guidance
- View livestock welfare schemes
- Manage animal healthcare records

The application uses a highly user-friendly interface with icons, multilingual support, and mobile responsiveness to make the system easy to use for rural users.

The platform also demonstrates the future potential of AI-powered veterinary healthcare systems in rural environments.

Problem Statement

Rural livestock owners face serious challenges in accessing veterinary healthcare facilities during emergencies.

The major problems identified are:

1. Lack of immediate veterinary support during emergencies.
2. Limited awareness regarding animal diseases and treatments.
3. Difficulty in locating nearby veterinary doctors.
4. Poor communication infrastructure in villages.
5. Lack of digital healthcare systems for livestock.
6. Inadequate information about government livestock schemes.
7. Absence of offline healthcare guidance systems.
8. Low accessibility of veterinary knowledge.
9. Delay in treatment causing animal deaths.
10. Poor management of livestock health records.

These problems reduce livestock productivity and increase economic losses for farmers.

Therefore, there is a need for a centralized, digital, mobile-friendly veterinary healthcare platform specially designed for rural communities.

Existing System

The traditional veterinary healthcare system in rural areas mainly depends on physical veterinary hospitals, government healthcare camps, and manual consultation processes.

In the existing system:

- Farmers must physically travel to veterinary hospitals.
- Emergency services are difficult to access.
- There is no centralized online veterinary support system.
- First aid information is not easily available.
- Government scheme awareness is limited.
- Veterinary consultations are mostly manual.
- Healthcare records are not digitally maintained.

The existing system lacks digital transformation and smart healthcare accessibility.

Most rural farmers rely on:

- Local informal advice
- Traditional methods
- Delayed veterinary visits
- Word-of-mouth healthcare guidance

This creates healthcare inefficiency and delays critical treatment.

Drawbacks of Existing System

The existing veterinary healthcare system has several limitations.

1. Lack of Emergency Accessibility

Farmers cannot quickly contact veterinary doctors during emergencies.

2. Poor Awareness

Many livestock owners do not understand disease symptoms and preventive healthcare.

3. No Real-Time Support

Traditional systems do not provide 24/7 support.

4. Transportation Issues

Rural villages may be far from veterinary hospitals.

5. Limited First Aid Knowledge

Farmers may not know basic first aid procedures for animals.

6. No AI Assistance

Existing systems do not provide intelligent healthcare guidance.

7. Poor Digital Infrastructure

Manual record management leads to inefficiency.

8. Language Barriers

Many platforms are not designed for local language support.

9. Lack of Mobile Optimization

Traditional systems are not optimized for smartphones.

10. Government Scheme Unawareness

Farmers often miss subsidies and benefits due to lack of awareness.

Proposed System

GORAKSHA – Pashu Sanjeevini is proposed as an integrated smart veterinary healthcare platform.

The proposed system includes:

- Emergency support system
- Veterinary doctor finder
- AI-powered healthcare chatbot
- Offline first aid guidance
- Government scheme awareness portal
- User profile management
- Responsive mobile-friendly interface
- Search and filter functionality
- Real-time interactive support

The system focuses on simplicity, accessibility, and practical usability.

The proposed platform can function as:

- Web application
- Progressive Web App (PWA)
- Future mobile application
- Rural healthcare support system

The system is specially designed to help farmers during emergency situations and provide immediate guidance before professional veterinary assistance arrives.

Objectives

The major objectives of the project are:

1. To provide quick veterinary healthcare access.
2. To support farmers during animal emergencies.
3. To improve rural veterinary awareness.
4. To integrate AI into livestock healthcare systems.
5. To provide offline first aid support.
6. To help users locate nearby veterinary doctors.
7. To improve government scheme awareness.
8. To design a mobile-responsive interface.
9. To create a farmer-friendly user experience.
10. To reduce livestock mortality due to delayed treatment.
11. To improve communication between farmers and veterinary doctors.
12. To develop a scalable future-ready healthcare platform.

Scope of the Project

The scope of GORAKSHA – Pashu Sanjeevini is very wide and can expand significantly in future.

Current Scope

- Emergency veterinary support
- Doctor search system
- First aid healthcare guidance
- AI chatbot interaction
- Government livestock scheme information
- User healthcare profile
- Mobile-friendly web application

Future Scope

- Real GPS-based tracking
- Live video consultation
- AI disease prediction
- Voice-based farmer assistance
- Multi-language support

- Vaccination reminder systems
- IoT sensor integration
- Cloud database connectivity
- Telemedicine platform
- Real-time ambulance support

The project can become a complete digital veterinary ecosystem for rural India.

Technologies Used

1. HTML5

HTML5 is used to structure the web application.

It defines:

- Navigation system
- Page layouts
- Forms
- Sections
- Buttons
- Containers
- Interactive elements

HTML provides semantic structure and accessibility.

2. CSS3

CSS3 is used for designing and styling the user interface.

CSS features include:

- Responsive layouts
- Animations
- Gradients
- Shadows
- Hover effects
- Mobile optimization
- Flexbox
- Grid system

- Typography

CSS improves the visual quality and user experience.

3. JavaScript

JavaScript provides dynamic functionality.

JavaScript handles:

- Page navigation
- Dynamic rendering
- Search filtering
- Modal popups
- Chatbot communication
- User interactions
- Real-time updates
- View switching
- Event handling

4. Claude AI API

Claude AI API is integrated for intelligent chatbot assistance.

The AI system helps users:

- Understand symptoms
- Get first aid guidance
- Learn preventive healthcare
- Receive healthcare suggestions
- Access conversational support

5. Responsive Web Design

The project uses responsive web design principles to support:

- Mobile phones
- Tablets
- Desktop systems
- Low-resolution devices

6. Frontend Architecture

The application follows a frontend-only architecture using:

- Component-based sections
- Dynamic rendering
- State-based interactions
- Client-side execution

HTML, CSS, JavaScript Explanation

HTML Structure

The HTML portion organizes the platform into multiple pages and sections.

Main HTML sections include:

- Navigation bar
- Hero section
- Doctor module
- First aid module
- Scheme module
- Profile module
- Chatbot system
- Emergency modals
- Responsive bottom navigation

The HTML structure uses semantic organization for better readability and maintainability.

CSS Design Principles

The CSS architecture focuses on:

- Visual consistency
- Accessibility
- Mobile responsiveness
- Smooth animations
- Modern healthcare design
- Rural-friendly readability

Important CSS Features

1. CSS Variables

The project uses reusable CSS variables for:

- Colors
- Shadows
- Border radius
- Theme consistency

2. Responsive Design

Media queries optimize the interface for smaller devices.

3. Animations

Animations improve user engagement through:

- Floating icons
- Pulse effects
- Fade transitions
- Hover interactions

4. Layout Systems

The system uses:

- Flexbox
- CSS Grid
- Adaptive spacing
- Dynamic alignment

JavaScript Functionality

JavaScript acts as the core interactive engine of the platform.

Dynamic Features Controlled by JavaScript

1. Doctor filtering
2. Search functionality
3. Emergency call simulation
4. Modal management
5. AI chatbot communication
6. Dynamic rendering of healthcare content
7. Navigation switching
8. Toast notifications

9. Map view simulation
10. Offline healthcare content display

Event Handling

JavaScript handles user actions such as:

- Click events
- Input changes
- Search operations
- Chat interactions
- View toggling

DOM Manipulation

The system dynamically updates webpage content using:

- innerHTML rendering
- Dynamic card generation
- Real-time interface updates

System Architecture

The project follows a client-side modular architecture.

Main Components

1. User Interface Layer

This layer handles visual interaction.

Components include:

- Navigation
- Forms
- Buttons
- Cards
- Modals
- Chat window

2. Functional Logic Layer

Implemented using JavaScript.

Responsibilities include:

- Data processing
- Search filtering
- State management
- Dynamic rendering
- Event handling

3. AI Integration Layer

The Claude API acts as an intelligent healthcare assistant.

Functions include:

- Conversational interaction
- AI healthcare responses
- Symptom assistance

4. Data Layer

Static JavaScript arrays are currently used for:

- Doctor data
- Scheme information
- First aid data

Future versions can integrate:

- Cloud databases
- Firebase
- MongoDB
- MySQL

Workflow Diagram Explanation

The workflow of the system follows a user-centered interaction process.

Workflow Steps

1. User opens the application.
2. Navigation system loads modules.
3. User selects required functionality.
4. JavaScript handles interaction.
5. Dynamic content is displayed.
6. AI chatbot responds to queries.
7. Emergency system assists users.
8. Doctor finder displays nearby doctors.
9. First aid guidance is shown.
10. Government schemes are displayed.

The workflow focuses on reducing complexity and improving accessibility.

Modules Description

The project contains several major modules.

Main Modules

1. Home Dashboard Module
2. Emergency Support Module
3. Doctor Finder Module
4. First Aid Module
5. AI Chatbot Module
6. Government Schemes Module
7. User Profile Module
8. Responsive Navigation Module
9. Search and Filter Module
10. Notification and Interaction Module

Each module is independently designed but integrated into a unified healthcare platform.

Emergency Support Module

The Emergency Support Module is one of the most important features of the project.

Purpose

To provide immediate veterinary assistance during emergencies.

Features

- Emergency button
- Veterinary doctor connection simulation
- Quick access support
- Real-time emergency interface
- Interactive call modal

Working Process

1. User clicks emergency button.
2. Emergency modal opens.
3. Nearby doctor details are displayed.
4. Simulated call connection begins.
5. User receives guidance.

Importance

This module helps reduce delays during critical situations such as:

- Animal poisoning
- High fever
- Injuries
- Delivery complications
- Severe infections

UI Design

The emergency button uses:

- Red color coding
- Pulse animation
- Large button size
- Clear visibility

This ensures immediate attention during stress situations.

Doctor Finder Module

The Doctor Finder Module helps users locate nearby veterinary doctors.

Features

- Doctor listing
- Search functionality
- Distance display
- Rating system
- Government/private categorization
- Map view simulation
- Call buttons

Search System

Users can search doctors based on:

- Name
- Village
- Specialization
- Location

Filter Features

The module supports:

- Nearest doctors
- Government doctors
- Private doctors
- Top-rated doctors

Benefits

- Faster veterinary access
- Improved healthcare communication
- Better emergency management
- Easy doctor discovery

Technical Implementation

JavaScript dynamically renders doctor cards from data arrays.

Filtering is performed using:

- Array filtering
- Sorting algorithms
- Dynamic rendering methods

AI Chatbot Module

The AI Chatbot Module provides intelligent healthcare assistance.

Purpose

To help farmers receive quick healthcare guidance.

AI Integration

The chatbot uses Claude API for conversational AI responses.

Features

- Symptom guidance
- Healthcare advice
- Vaccination information
- Scheme guidance
- Multilingual interaction
- Quick response buttons

Working Process

1. User enters question.
2. JavaScript sends API request.
3. Claude AI processes input.
4. Response is displayed dynamically.

Benefits

- 24/7 assistance
- Improved awareness
- Reduced confusion
- Faster information access

Rural Usability

The chatbot is designed with:

- Simple English
- Kannada phrases
- Short responses
- Easy readability

This makes the system more accessible for rural users.

First Aid Module

The First Aid Module provides emergency healthcare instructions.

Supported Conditions

- Fever
- Injury
- Infection
- Digestion issues
- Pregnancy complications
- Poisoning

Features

- Step-by-step guidance
- Warning alerts
- Educational videos
- Offline support
- Emergency instructions

Offline Capability

The module is preloaded into the system.

Users can access healthcare guidance without internet.

Importance

This module is extremely useful in remote villages where:

- Internet access is poor
- Doctors are unavailable
- Emergency response is delayed

Technical Structure

First aid data is stored in structured JavaScript objects.

Dynamic rendering allows:

- Fast loading
- Easy updates
- Interactive switching

Government Schemes Module

This module improves awareness regarding livestock welfare schemes.

Purpose

To educate farmers about government support programs.

Features

- Scheme cards
- Funding information
- Benefit descriptions
- Apply buttons
- Awareness campaigns

Included Schemes

- Rashtriya Gokul Mission
- Livestock Health & Disease Control
- PM-KISAN Animal Link
- Kisan Credit Card
- Pashu Sanjivani Scheme
- Mobile Veterinary Units

Benefits

- Financial support awareness
- Improved livestock investment
- Better rural development
- Government connectivity

UI Design

Colorful cards and icons improve readability and engagement.

User Profile Module

The User Profile Module stores farmer and livestock information.

Features

- Farmer details
- Animal count
- Vaccination records
- Consultation history
- Language preferences
- Settings management

Advantages

- Better healthcare tracking
- Record management
- Vaccination monitoring

- Personalized experience

Future Improvements

Future versions may include:

- Cloud synchronization
- Health history analytics
- IoT integration
- QR-based livestock identification

Responsive UI Design

Responsive design is a major strength of the project.

Mobile-First Approach

The platform is optimized primarily for smartphones.

This is important because rural users mostly use mobile devices.

Responsive Features

- Adaptive layouts
- Flexible grids
- Bottom navigation
- Mobile-friendly buttons
- Scalable typography
- Touch optimization

CSS Techniques Used

- Media queries
- Flexbox
- Grid layouts
- Relative sizing
- Dynamic spacing

Benefits

- Better usability
- Improved accessibility
- Faster interaction
- Cross-device compatibility

Offline Support System

The project provides offline-friendly healthcare guidance.

Importance

Many rural areas experience:

- Poor internet connectivity
- Network interruptions
- Remote locations

Offline support ensures healthcare accessibility during such situations.

Offline Components

- First aid instructions
- Healthcare guidance
- Preloaded emergency information

Future Improvements

Future versions can use:

- Progressive Web Apps (PWA)
- Service Workers
- Local storage caching
- Offline databases

Search and Filter Functionality

The search and filter system improves accessibility and usability.

Search Features

Users can search:

- Doctors
- Locations
- Specializations

Filtering Features

The system supports:

- Government doctors
- Private doctors
- Distance sorting
- Rating sorting

Technical Working

JavaScript array methods are used:

- filter()
- sort()
- includes()

Advantages

- Faster information access
- Reduced user effort
- Better navigation experience

Real-time Interaction Features

The project includes several interactive components.

Real-Time Elements

- Toast notifications
- Dynamic rendering
- Chatbot responses
- Interactive modals
- Hover animations
- Live filtering

User Experience Benefits

- Better engagement
- Faster interaction
- Improved accessibility
- Modern application feel

JavaScript Role

JavaScript continuously updates the user interface without reloading the webpage.

Advantages

GORAKSHA – Pashu Sanjeevini provides multiple advantages.

Major Advantages

1. Easy veterinary access
2. Rural-friendly interface
3. Emergency healthcare support
4. AI-powered guidance
5. Offline first aid system
6. Mobile responsiveness
7. Government scheme awareness
8. Faster healthcare communication
9. Improved livestock management
10. Future scalability

Social Impact

The project can significantly improve:

- Rural healthcare awareness
- Livestock productivity
- Farmer confidence
- Animal survival rates

Limitations

Although the project is highly useful, some limitations exist.

Current Limitations

1. No backend database
2. Simulated doctor data
3. No real GPS integration
4. No real-time video consultation
5. No authentication system
6. Internet dependency for AI chatbot
7. No cloud synchronization

Technical Constraints

The current version is a frontend prototype system.

Future development can overcome these limitations.

Applications

The project has several real-world applications.

Main Applications

- Rural veterinary healthcare
- Livestock emergency support
- Animal healthcare awareness

- Farmer assistance systems
- Veterinary telemedicine
- Government livestock outreach
- Digital agriculture platforms

Deployment Possibilities

The project can be deployed as:

- Web application
- Android application
- Progressive Web App
- Government healthcare platform

Future Enhancements

The future scope of the project is very large.

Proposed Enhancements

1. Real GPS Tracking

Integration with Google Maps for live doctor locations.

2. Video Consultation

Real-time veterinary video calling.

3. AI Disease Prediction

Machine learning models for symptom analysis.

4. Voice Assistant

Voice-based rural healthcare guidance.

5. Multi-language Support

Kannada, Hindi, Marathi, Telugu, Tamil, and other languages.

6. Vaccination Alerts

Automated reminders for animal healthcare.

7. Cloud Database

Cloud storage for healthcare records.

8. IoT Integration

Smart sensors for:

- Temperature monitoring
- Heart rate monitoring
- Animal activity tracking

9. Emergency Ambulance System

Live veterinary emergency transport support.

10. AI Vision System

Image-based disease identification.

Security Improvements

Security is important for healthcare applications.

Recommended Security Features

1. User authentication
2. Secure API keys
3. HTTPS encryption
4. Database security
5. Access control
6. Cloud security
7. Secure communication protocols

Claude API Security

API keys should be stored securely using:

- Backend servers
- Environment variables
- Token protection systems

Performance Optimization

Performance optimization improves application speed and efficiency.

Optimization Techniques

1. Minification

Compressing CSS and JavaScript files.

2. Lazy Loading

Loading resources only when required.

3. Image Compression

Reducing image size for faster loading.

4. Efficient Rendering

Optimized DOM updates.

5. Caching

Using local storage and browser caching.

6. API Optimization

Reducing unnecessary API requests.

Benefits

- Faster loading
- Better responsiveness

- Improved user experience
- Lower bandwidth usage

Real-world Use Cases

The project can solve many practical rural healthcare problems.

Use Case 1: Emergency Fever

A farmer notices high fever in a cow.

The platform provides:

- First aid guidance
- Emergency doctor contact
- AI symptom assistance

Use Case 2: Animal Injury

A goat gets injured during grazing.

The farmer can:

- Access wound treatment guidance
- Contact nearby doctors
- Learn infection prevention methods

Use Case 3: Pregnancy Support

A buffalo experiences delivery complications.

The application helps with:

- Labour monitoring guidance
- Emergency assistance
- Veterinary contact

Use Case 4: Government Subsidy Awareness

A farmer wants livestock subsidy information.

The schemes module provides:

- Funding details
- Awareness information
- Application guidance

Testing and Results

The project was tested for:

- Responsiveness
- Functionality
- Navigation
- Dynamic rendering
- Search operations
- Chatbot integration
- UI consistency

Testing Methods

1. Functional Testing

Verified all features individually.

2. Responsive Testing

Tested on:

- Mobile screens
- Tablets
- Desktop devices

3. Interaction Testing

Verified:

- Buttons
- Modals
- Animations

- Navigation

4. Search Testing

Validated filtering and search operations.

5. API Testing

Verified chatbot API communication.

Results

The project successfully demonstrated:

- Smooth navigation
- Interactive functionality
- Responsive design
- Dynamic user experience
- Healthcare accessibility

The platform proved highly suitable for rural veterinary assistance.

Conclusion

GORAKSHA – Pashu Sanjeevini is an innovative rural animal healthcare platform developed to improve veterinary accessibility and farmer support.

The project successfully combines:

- Modern web technologies
- AI integration
- Emergency healthcare systems
- Mobile-responsive design
- Rural usability principles

The application demonstrates how digital healthcare systems can improve livestock welfare and reduce healthcare delays in rural communities.

The platform provides a strong foundation for future development into a full-scale smart veterinary healthcare ecosystem.

By integrating AI, responsive design, offline support, and healthcare awareness, the project contributes significantly toward digital rural transformation.

The project also highlights the importance of user-centered design in healthcare technology.

GORAKSHA – Pashu Sanjeevini has the potential to become a valuable tool for farmers, veterinary professionals, government livestock departments, and rural healthcare organizations.

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Final Summary

GORAKSHA – Pashu Sanjeevini is more than a simple web application. It is a smart rural healthcare initiative focused on improving livestock welfare through technology.

The platform demonstrates:

- Technical innovation
- Social impact
- Practical usability
- AI integration
- Emergency healthcare accessibility

The project successfully addresses real-world rural problems and presents a scalable solution for the future of veterinary healthcare systems.

The combination of emergency support, AI chatbot guidance, first aid education, government awareness, and responsive design makes the project highly valuable for modern rural digital transformation.